

WICAKSONO LEKSONO MUHAMAD

+62 878 8761 7782 • wlap.atacama@gmail.com

linkedin.com/in/wicaksonolxn • GitHub: github.com/wicaksonoleksono, github.com/airlanggawicaksono • wicaksonoleksono.github.io

AI/ML Engineer · NLP Researcher · Data Scientist

PROFESSIONAL SUMMARY

AI Software Engineer focused on LLM systems, NLP, and ML deployment. Built production AI applications using LangGraph, RAG, PyTorch, Docker, and Kubernetes. First-author on two ACL-affiliated shared-task papers, with deployed systems in education, healthcare, and climate-policy use cases.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Go, SQL, Dart

ML & NLP: PyTorch, HuggingFace Transformers, scikit-learn, fine-tuning, contrastive learning, knowledge distillation, model optimization, embeddings, mechanistic interpretability

LLM Systems: LangGraph, LangChain, LangSmith, RAG, OpenAI API, ChromaDB, MCP, prompt engineering

Backend & Infra: Flask, Next.js, REST API, gRPC, Docker, Kubernetes, nginx, Linux, CI/CD, GCP, AWS

EXPERIENCE

Researcher and Lead Software Engineer

Feb. 2022 – Present

Mantera Studio Research Lab (PT. BSM)

Jakarta, Indonesia

- Published first-author at SemEval-2026 Task 11, finishing top-5 across all subtasks on multilingual formal reasoning via deterministic parsing and structural abstraction of syllogisms into canonical logical forms.
- Published first-author at BEA 2026 Shared Task, ranking top-15 across all L1 groups (14th Spanish, 11th German, 7th Mandarin) with a Ridge ensemble over XLM-RoBERTa and mDeBERTa using ordinal soft contrastive learning.
- Lead engineering across the lab's production AI portfolio, directing system architecture and shipping customer-facing LLM applications in collaboration with Queen's University Belfast, UGM, and Deakin University.

Research Assistant, Machine Learning Systems

Feb. 2026 – Present

Universitas Gadjah Mada

Yogyakarta, Indonesia

- Developing early-exit training and inference methods for energy-efficient, low-latency deployment on NVIDIA Jetson Orin across 7W–25W power budgets.

Research Assistant, Software Engineering Lab

Mar. 2025 – Oct. 2025

Universitas Gadjah Mada

Yogyakarta, Indonesia

- Led ML workstream for a national grant: delivered PHQ-aligned LLM screening with deep-learning facial affect analysis, evaluated on 302 samples, and presented results to university stakeholders and grant reviewers.
- Built LLM-based NLP tooling for requirements engineering, adding pragmatic and semantic quality checks for user-story specifications that extended prior automated QUS analysis.

Web Developer

2019 – 2022

Mastej Studio

Malang, Indonesia

- Built and shipped What's On Malang, a React/Vite tourism platform (400+ DAU, 100 leads/day), and the Begawan apartment marketing site with 360-degree virtual tours (10 daily inquiries); delivered demos and prototypes directly to regional clients.
URLs: whatsonmalang.com · begawan-apartment.com
- Co-developed Janur, a digital wedding-invitation site with built-in attendance tracking. Helping build 7 web based custom invitations for weddings
URL: janur.wedding
- Supported Pasar Santai, an annual Ramadan local-market event in Malang, as logistics officer and web developer.
URL: pasarsantai.themastej.com

PUBLICATIONS

ITLC at SemEval-2026 Task 11: Normalization and Deterministic Parsing for Formal Reasoning in LLMs. First author (Mantera Studio), SemEval-2026 (ACL). Top-5 across all subtasks, outperforming fine-tuning and activation-level baselines.

URL : arxiv.org/abs/2603.02676

Improving Lexical Difficulty Prediction with Context-Aligned Contrastive Learning and Ridge Ensembling. First author (Mantera Studio), BEA 2026 Shared Task (ACL). Top-15 across all L1 groups (14th ES, 11th DE, 7th ZH).

URL : arxiv.org/abs/2605.08950

Aligning Implied Statements for Implicit Hate Speech Generalizability. First author, under review. +4–8% out-of-domain F1 across 3 benchmarks via triplet contrastive learning with context-bounded semi-hard negatives.

URL : openreview.net/pdf?id=pgwGGyy77k

On-Edge Device Optimization Using Multiple Classification Method for a Cat and Dog Audio Classifier. First author, IEEE. 95.2% accuracy, 3 ms latency, 3.8 KB memory on Arduino Nano 33 BLE.
URL : ieeexplore.ieee.org/abstract/document/10933173

Enhancing Automated User Story Quality Assessment with Large Language Models. Co-author, ICODSE 2025.
URL : <https://ieeexplore.ieee.org/abstract/document/11351886>

SELECTED PROJECTS

- Dyslexic Reader** (Mantera Studio × UGM × Queen’s University Belfast) · *Dart, Next.js, Kubernetes, Python, gRPC, webhook* 2026
URL : dyslexic.app
- Built LLM-powered dyslexia support app for a Mantera Studio collaboration with Queen’s University Belfast and UGM, deployed on Kubernetes with Dart mobile and Next.js web frontends across multiple VPS.
 - Implemented a real-time font-rendering dyslexia lens, LLM summarization and text-professionalization, and an AHRQ-aligned screening protocol for clinical alignment.
- Mental Health Screening Platform** (Mantera Studio) · *Flask, LangGraph, LangChain, PyTorch,webhook,LiteRT* Nov. 2025
URL : mentalhealth.laptopmerahputih.id
- Shipped LLM-driven PHQ + CPAS screening workflows with LangGraph agentic orchestration, serving 40+ concurrent sessions with stable uptime.
 - Integrated deep-learning facial affect capture; cut p95 latency and serving cost 90% via local response batching and request queuing, monitored with LangSmith.
- RAG Chatbots: TASYA & IndoClimate** (Mantera Studio × UGM, Deakin) · *Flask, React, ChromaDB, LangGraph* 2025
URLs : tasya.dtmiumgm.id · chat.indoclimate.id
- Built WhatsApp webhook and REST RAG services over institutional docs and Indonesian climate legislation; indexed ~2,000 pages across 7 documents into ChromaDB.
 - Orchestrated multi-step retrieval and generation via LangGraph; added an LLM summarization pipeline for long legal documents to improve vector matching, traced with LangSmith.
- SOC Dashboard** (Mantera Studio · under NDA) · *Vite, Redux Toolkit, Go, LangGraph, LangSmith* 2024
- Refactored frontend to domain-driven design, cutting redundant code 70%; integrated server telemetry ingestion, distributed tracing, audit logging, and LLM-based log summarization.
- Sijinak Attendance System** (Mantera Studio) · *gRPC, FFI, Dart, Python, C* 2025
URL : github.com/airlangawicaksono/sijinak_win, <http://sim.man2yogyakarta.sch.id/>
- Built RFID-based attendance system integrating Hikvision hardware via Foreign Function Interface to a native C SDK for hardware interaction of the RFID device used; exposed a microservice layer over gRPC with a Dart frontend.

EDUCATION

Universitas Gadjah Mada <i>Master of Artificial Intelligence</i>	Yogyakarta, Indonesia <i>Feb. 2024 – Oct. 2025</i>
Institut Teknologi Sepuluh Nopember <i>B.A.Sc. Computational and Applied Mathematics</i>	Surabaya, Indonesia <i>Aug. 2018 – Nov. 2022</i>